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ABSTRACT

Clear air turbulence (CAT) has been a problem to aircraft designers and pilots since the earliest days of manned flight. However many years elapsed before techniques were developed to give quantitative measurements of gust velocities which enabled the aircraft designer to give structural design consideration to the stresses imposed by rough air. An early method of mathematically defining atmospheric turbulence, still widely used, assumes each aircraft gust encounter to be a discrete, isolated occurrence. The modern power spectral density concept assumes atmospheric turbulence to be a continuous phenomenon and relates gust power to gust wavelength from which frequency inputs can be determined and aircraft resonance or "tuning" (causing catastrophic loads) can be predicted. Large, flexible high speed aerospace vehicles may experience large load inputs from long wavelength gusts which have little effect on slower, more rigid contemporary aircraft. Therefore new gust probes, sensors, test vehicles, etc. are under development. A typical airborne turbulence system capable of continuous true gust velocity measurements records at least twenty variables on magnetic tape. Systems of this type are currently installed in various AF aircraft as part of an overall USAF program to define the atmospheric environment from sea level to 200,000 ft.

INTRODUCTION

Past and present day aircraft have encountered clear air turbulence (CAT) from sea level to 70,000 ft. Meteorological investigations indicate the presence of CAT up to the 200,000 ft. level. Therefore, it seems there is no altitude level within

the atmosphere layer used by winged aircraft that is completely devoid of CAT, not even at the operational heights of the Lockheed U-2 or the North American X-15.

The two primary causes of CAT are (1) convective motions in an unstable atmosphere, and (2) perturbation motions in a stable atmosphere. At high altitudes another type of turbulence known as undulance (gravity waves) appears to exist which, under certain dynamic conditions, may cause higher structural loads than the more common random type of turbulence.

For many years the magnitude of turbulence could only be estimated by pilots from "seat of the pants" accelerations. Not until techniques were developed to give quantitative measurements of gust velocities, could the aircraft designer give structural design consideration to the stresses imposed by rough air.

One of the early methods of mathematically defining atmospheric turbulence, which is still in wide use today, assumes each aircraft gust encounter to be a discrete gust encounter. Three parameters, V (velocity), G (c.g. accelerations) and H (altitude), are recorded in-flight. From this information, combined with certain aircraft characteristics, derived gust (equivalent) velocities (U_{de}) in ft/sec. can be computed. These are not true gust velocities as might be measured by an anemometer but can be used to predict equivalent loads on aircraft of known response characteristics.

VGII data is used to establish frequency of occurrence of gusts for various altitudes, geographic locations and seasons. Structural design criteria based on these data have been used with success on standard configuration, comparatively slow and rigid aircraft of today. However, the trend is toward lower ratios of structural weight to gross weight for large aerospace vehicles resulting in relatively flexible structures. Present day gust criteria based on the discrete gust approach (no wavelength or frequency considered, only gust magnitude) cannot be used with confidence in structural design of future vehicles.

The more advanced power spectral density (psd) concept assumes atmospheric turbulence to be a continuous phenomenon and relates gust power to gust wavelength. Thus, when the velocity of the proposed vehicle is known frequency inputs can

be determined and aircraft resonance or "tuning" (which would cause catastrophic loads) can be predicted. Therefore, it is necessary to obtain continuous turbulence data, in psd format, of sufficient quality and quantity as required to establish adequate, rational structural design criteria for future aerospace vehicles.

BACKGROUND

Accident records maintained by both military and civilian agencies give ample testimony to the destructive power of turbulent air. In many cases, structural failure and/or loss of control have been the end results of severe gust encounters. Most of these accidents prior to World War II were due to low altitude turbulence induced by terrain characteristics. Not until after World War II when pressurized aircraft began penetrating the lower reaches of the jetstream, did a new phenomenon known as clear air turbulence (CAT) begin to appear.

The straight wing, supercharged piston aircraft which first used this high altitude regime had a wide range of operating speeds between stall and maximum cruise. This allowed the aircraft to be slowed down (normal recommended procedure) considerably in the event of a CAT encounter and still have an adequate speed margin above stall.

However, the modern subsonic sweptwing jet aircraft flies in a comparatively narrow speed zone which converges with altitude. The boundaries of this zone are formed by the low speed stall and high speed stall (buffet) characteristics, respectively. Thus, a jet aircraft cannot appreciably reduce speed in the event of a gust encounter without risking loss of control due to a stall.

This pilot's dilemma is graphically illustrated in a report of a CAT encounter by a United Air Lines Boeing 720 on 12 July 1963. During a period of only 90 seconds the aircraft was:

- a. Beyond the stall and high speed boundaries several times, and at least one time fully stalled.
- b. At minimum speeds of 213 knots (Mach 0.69) and maximum speeds beyond 470 knots indicated air speed (Mach 0.93).

- c. Subjected to peak g loads of +3.2 and -1.4 (that is the g loads varied between +2.2 and -2.4 of the normal 1 g).

Other airline aircraft have not been so fortunate as to recover from a loss of control as this one did. On 12 Feb 1963, a Northwest Airlines Boeing 720B crashed near Miami, Florida, killing all 43 aboard. It was presumably flying in clear air near a line of thunderstorms. Many other CAT incidents/accidents have been reported to the Federal Aviation Agency (FAA) causing that agency to initiate investigative programs to study various aspects of the problem.

During the 1961-1964 period, the Air Force reported 5 accidents and 37 incidents due to CAT. Seventeen fatalities and twenty non-fatal injuries and a dollar loss of thirty five million resulted from these accidents.

The true costs of clear air turbulence extend beyond injuries to humans or damage to aircraft; however. They range from reduction of fatigue life of the aircraft structure to lawsuits resulting from deaths, injuries and property damage. As the amount of high altitude traffic continues to mount, so will the losses due to CAT increase, unless adequate research work is done in three technical areas:

- a. Obtain continuous turbulence data to provide adequate structural design criteria.
- b. Improve meteorological forecast capability.
- c. Develop airborne CAT detection devices.

CURRENT GUST CRITERIA REQUIREMENTS

The aerospace industry is currently engaged in designing vehicles which will operate at all altitudes from sea level through the ozonosphere, flying at speeds ranging from 0 to supersonic. A partial listing of such vehicles would include:

- a. F-111 - tactical fighter designed to fly to Mach 2.5 at altitude.
- b. C-5A - Large logistics transport designed to carry outsize cargo from minimum airfields, operating at high subsonic speeds at low and high altitudes.
- c. C-142 V/STOL transport - Large tilt wing aircraft carrying 32 troops or 8000 lbs. cargo.

- d. SR 71/F-12A - high altitude, supersonic reconnaissance/interceptor aircraft.
- e. SST-Supersonic Transport designed to carry in excess of 200 passengers at MACH 3.

f. 707/DC-8 - enlarged versions of present day jet transport aircraft capable of carrying up to 275 passengers at high subsonic speeds.

All of these aircraft at various times will be flying at altitudes and geographic locations during seasons of peak turbulence activity. Some turbulence avoidance capability may be assumed for subsonic aircraft but not for high altitude supersonic aircraft. Therefore, adequate design criteria must be provided to insure structural integrity. To establish adequate gust criteria requirements for these varied, multi-purpose aircraft, it is necessary to obtain the following turbulence data:

- a. Gust wavelength measurements from 40 to 60,000 ft (1 ft/sec RMS)..
- b. Sufficient data sample size to insure statistical stationarity.
- c. Establish spectral shapes for various CAT phenomena at all altitudes.
- d. Compute probability density functions based on true gust velocities.
- e. Sufficient VGH data to plot frequency of occurrence curves (miles per gust, U_{de}).
- f. Sufficient VGH data to provide statistical data on turbulence associated with certain route structure, mission profiles, etc.

TYPES OF TURBULENCE PROGRAMS

a. Pilot Report - Man himself can be thought of as a complete gust measuring system. "Seat of the pants" accelerations are transmitted by proper circuitry and interpreted by the brain as light, moderate, severe or extreme turbulence. Since the manufacturing tolerances for human beings are unacceptably large from an engineering point of view, the estimation of gust intensity are too gross to be of engineering value.

However, many programs have been and continue to be conducted utilizing pilots' estimates of gust intensities which are in turn related to other weather phenomena, geographical location, etc. Prepared forms are filled out by the pilot and incorporated

into a statistical sample which may be categorized according to route structures, seasonal changes, altitude bands, etc. These data may be used to establish frequency of occurrence, verification of forecasts and other meteorological analysis.

b. VGH Programs - The fundamental relationship assumed in the derived gust velocity equation is that the intensity of an isolated gust is proportional to normal acceleration. Other factors of interest to the design engineer are gust shape and frequency of occurrence. The AWC-1 Panel designated the gust shape as $\frac{u}{U} \left(\frac{S}{H} \right) = 1/2 (1 - \cos \frac{S}{H})$. The value of (H) was designated equal to 12.5 chord lengths, (s) is the distance of penetration into the gust in chords, (u) is gust velocity at any penetration distance into the gust in ft/sec and (U) is the maximum value of the gust velocity. By varying the assumed gust shape and gust chord length, some aircraft dynamic response characteristics can be predicted.

Several simplifying assumptions are made in the derived gust velocity method as follows:

- (1) The airplane is a rigid body.
- (2) The airplane forward speed is constant.
- (3) The airplane can rise but not pitch.
- (4) The airplane is in steady level flight before entry into the gust.
- (5) The lift increments of the fuselage and horizontal tail are negligible in comparison with the wing lift increments.
- (6) The gust velocity is uniform across the span and parallel to the vertical axis of the aircraft at any instant.

The equation for calculation of derived gust velocity is:

$$U_{de} = \frac{\Delta H}{V_e P_o} \quad \frac{2W}{KgMS}$$

Where:

U_{de} - derived gust velocity, ft/sec

ΔN - normal acceleration at c.g., g

V_e - equivalent airspeed, ft/sec.

P_o - air density at sea level, slugs/cu. ft.

W - airplane wt., lb.

M - wing lift slope, per radian

S - wing area, sq. ft.

K_g - gust alleviation factor

The VGH parameters which are recorded in flight are $V_e, \Delta N$ and P_o (related to altitude), respectively. The VGH data is usually converted to gust velocity peak count. The number of derived gust velocity peaks are summed for magnitude bands for a given altitude band and time period.

Two aircraft dynamic response effects which lead to error in the derived gust velocity analysis are:

(1) accelerations caused by pilot input which cannot be precisely separated from accelerations caused by gust input, both of which are represented in the ΔN term.

(2) Effects of aircraft pitching (causing changes in accelerometer vector axis).

c. PSD Methods - The power spectral density method based on continuous turbulence measurements provides a more fundamental description of atmospheric turbulence than the discrete gust concept. Atmospheric turbulence may be defined as the variation of the velocity vector with time and position along a streamline of average wind flow. This variation appears to be random in both amplitude and wavelength; thus a statistical picture of the turbulence is required. Since both amplitudes and wavelengths are important, power spectral techniques are used to represent the turbulence, Figure 1. The total spectrum of atmospheric turbulence, which covers wavelengths (ft.) from zero to infinity, may be divided into roughly three wavelength regions as follows:

(1) Long wavelength region in which mechanical and convective energy enters the turbulent motions (Region I of Figure 1).

(2) Medium wavelength region in which the energy is transferred with significant gain or loss (Region II).

(3) Short wavelength region where viscous effects dissipate the energy (Region III).

Experimental data are in general agreement that the slope in Region II varies between $-5/3$ and -2 . Due to low power levels the spectral shape in Region III is largely underdefined and very little effort has been devoted to research in this area. However, Region I is of particular interest to the structural designer due to the high power levels which may cause large dynamic load inputs. Emphasis is now being placed on developing instrumentation and analysis technique for defining the wave shape in this region.

Various equations have been developed to define the gust spectral shape. One of the most popular expressions introduced and used by NASA is as follows:

$$\Phi_w(\Omega) = \frac{\sigma_w^2 L}{\pi} \cdot \frac{1 + 3I^2 \Omega^2}{(1 + \Omega^2 L^2)^2}$$

Where L is the turbulence scale parameter in ft. and Ω is the space frequency or wave number expressed as circular frequency. The equation has the practical advantage of being simple with a correspondingly simple correlation function. The turbulence scale parameter L often used with this expression is 1000 ft. except for altitudes of 200 - 300 ft. or less. Improved instrumentations techniques have, however, given experimental evidence that L is considerably longer. It has also been shown that this equation does not give a very good fit to the experimental data.

An equation developed by Von Karman (1948) has "fitted" very well with measured spectra and correlation function. This equation for one dimensional spectrum of the vertical component of turbulence is as follows:

$$G(\Omega) = \sigma_W^2 \frac{L}{\pi} \frac{1 + \frac{8}{3} (1.339 L \Omega)^2}{[1 + (1.339 L \Omega)^2]^{11/6}}$$

Unfortunately the mathematics of this equation and its correlation function are quite complicated.

Corresponding equations have been developed to convert measured in-flight data into power spectral density format and then compared with the foregoing theoretical expression. The power spectral density $G(\omega)$ is obtained from the time history of the random gust by an averaging process in the frequency domain:

$$G(\omega) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T [Y(t) e^{-i\omega t}]^2 dt$$

The process consist of filtering (using electrical analog or numerical filters) the time history to extract a selected frequency band, squaring the input quantity (root mean square) in this region and integrating to obtain the average. This provides a numerical distribution of the gust power as a function of frequency (wavelength).

The power spectral density function is directly related through the Fourier Transform to the auto covariance function. These function analyze the rate of change of the random input. The equations for the power spectral density function and the related auto covariance function respectively are as follows:

$$\begin{aligned} \Phi_x(\omega) &= 2\pi \int_0^\infty R_x(\tau) \cos \omega \tau d\tau \\ R_x(\tau) &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T}^T x(t) x(t+\tau) dt \end{aligned}$$

where $\Phi_x(\omega)$ is the power spectrum of a disturbance $x(t)$ and $R_x(\tau)$ is the auto covariance function is related to the auto correlation function by the ratio $\frac{R(\tau)}{R(0)}$ where (τ) is the time interval between samples usually about .05 seconds.

The power spectral density function and the auto correlation function completely describe the random sample if the statistical process has a Gaussian distribution.

Auto correlation functions define the characteristic of a single, continuous random sample.

Cross spectra are computed from two different random signals and the relationship between the two is established. Typical cross spectra for example may relate wing beam stress to c.g. accelerations.

d. True Gust Velocity Equations

The heart of any gust measuring system is the gust probe, Figure 2, usually mounted on a gust boom of appropriate length (usually 1 1/2 fuselage diameters). If the probe moves relative to a fixed spatial reference and also relative to the surrounding air mass, the velocity of the air mass relative to a fixed spatial reference (gust velocity) can be found by subtracting the velocity of the probe relative to the air mass from the velocity of the probe relative to the fixed reference. This simple concept forms the bases for a method of measuring atmospheric turbulence. The gust velocities thus obtained are true gust velocities and are free of pilot input and aircraft response characteristics.

Figure 3 presents the nomenclature for determining the equation for computation of vertical gust velocity and the true gust velocity equation. The derivation is as follows:

Let U_v = vertical air velocity
relative to space

U_n = vertical velocity of
probe in space

V_t = true airspeed of probe

θ , α , and γ are respectively the pitch attitude, angle of attack, and flight path angle of the probe. The instantaneous vertical velocity of the probe relative to the air mass is:

$$c = \gamma V_T \text{ (for small } \gamma \text{)}$$

The velocity of the probe relative to space is V_n as stated. Hence, the velocity of the air mass relative to space is

$$U_v = V_n - V_t$$

$$\text{but } \gamma = \alpha - \alpha$$

$$\text{hence } U_v = V_n - V_t \text{ (} \alpha - \alpha \text{)}$$

A fully instrumented Differential Pressure Gust Probe, Figure 2 (a), contains transducers for the measurement of pitch and yaw angles and q (dynamic pressure) from which α and β can be determined. Approximations to V_t can be determined from the measured values of q .

Two types of vane probes are also used in gust measuring work and can be described as follows:

(1) The movable vane type which has the vane mounted on low friction bearings allowing the vane to align itself with the relative air. Angular movements of the vane are recorded by a potentiometer.

(2) The fixed vane type, Figure 2 (b), which has the vane rigidly attached to a slotted beam. Flexures of the beam due to aerodynamic loads are measured by semiconductor strain gages and related to angle of attack.

These probes usually include two accelerometers, one each for the pitch and yaw planes, whose outputs, when integrated, give the normal and lateral velocities of the probe due to aircraft motions.

With the addition of a vertical gyro for the measurement of Θ , all the variables in the true vertical gust velocity equation can be measured. The true lateral gust velocity can be determined by measuring and analyzing essentially the same parameters in the lateral plane.

GUST DATA AVAILABLE - VGH

NASA pioneered the development and use of the discrete gust approach. Typical of published data available in this form are listed in chronological order as follows:

- a. "A review of NASA High Altitude Clear Air Turbulence Sampling Programs" Roy Steiner, AIAA Aerospace Science Meeting January 25 - 27 1965, New York, New York.
- b. The B-52 VGH Programs, James D. Yost, CONEA Structures Note 64-4, October 1964.
- c. "Turbulence and Precipitation Problems Associated with Operation of Supersonic Transport", Richard Rhyne and Roy Steiner, Fourth Conference on Applied Meteorology, September 10-14, 1962, Hampton, Virginia.
- d. "The Spectrum of Turbulence for Aircraft Fatigue Analyses," Lt. E. Joseph, July 1959, WPAFB, Ohio.

The NASA High Altitude Clear Air Turbulence Sampling Program report by Roy Steiner, (a) above, contains an analysis of approximately 800,000 flight miles of VGH data collected by operational AF Lockheed U-2 aircraft flying at altitudes of 40-70,000 feet. Four geographical areas of operations are identified but no meteorological information is given.

The B-52 VGH Programs reported by James D. Yost, (b) above, were designed to provide continuous monitoring of the structural loads encountered by operational aircraft. Interim service load programs (VGH) for fatigue damage calculations on the B-52 are:

- a. The Castle-Walker Program was conducted at Castle AFB and Walker AFB from October 1958 to January 1960. Approximately 1086 hours of VGH data was collected on six (6) B-52 B-F aircraft.
- b. The Heavy Gross Weight Program was conducted at five (5) Air Force bases from October 1960 to September 1961. Approximately 15,000 hours of VGH data was collected on twenty-seven (27) B-52B aircraft oscillographs. Of these data 1538 hours were reduced.

c. The B-52 B-F Service Loads Program (BCP 1019) was conducted at 11 Air Force bases from 1 February 1962 to 31 January 1964. Approximately 50,000 hours of VGH data and about 6000 hours of stress data was collected on 67 aircraft. About 13,000 hours of VGH data and 1400 hours of stress data was reduced.

d. The B-52 G-H Service Loads Program (BCP 1050 VGH) was conducted at six Air Force bases from 30 May 1963 to 31 May 1964. Seventeen thousand (17,000) hours of VGH data and five thousand (5,000) hours of stress data was collected on fifty-four aircraft. Eighty-six hundred (8600) hours of VGH data and two thousand hours of stress data was reduced.

e. The B-52 B-H VGH Program (BCP 1102) was conducted at eight (8) Air Force bases from 1 Feb 1964 to 31 July 1964.

f. The present B-52 VGH Recording Flight Loads History Program is being programmed for 25 Air Force bases from 1 August 1964 to 30 June 1965.

All of these Interim VGH Programs used the Century Type 409 oscillograph recorders.

In addition to these published data, similar programs are currently being conducted by NASA, USAF, FAA, etc. However, the inherent weaknesses in the basic data cannot be overcome. This view is supported in FAA Working Paper 201 titled "US Comments on Needs for Atmosphere Research and Current Programs", prepared in the Second French-Anglo-US-SST Meeting February 1-5, 1965. Recommendation 5.2.2 - Turbulence Data - reads as follows:

"A reassessment of design needs for turbulence data has been made. We believe that we have a sufficient amount of derived gust data in U_{de} form. The fact that we do not believe that meaningful design information either for static or fatigue loading conditions can be readily derived from discrete gust data is relevant to this assessment of U_{de} data. More data in power spectral form is needed."

Working Paper 202 prepared by the same agency states, under Item C 2 - Transient Stresses, the following:

"The supersonic transport will be more flexible than current transports and will have different relationships between elastic and rigid body frequencies. These differences can result in critical dynamic response to gust and taxi loading conditions. The more rational methods developed in the last decade to assess the response of aircraft to the gust and taxi conditions encountered during aircraft operations should be used for the SST to assure an adequate level of safety. Research on swept and unswept airplanes and service experience on both civil and military aircraft indicates that it is essential to take into account the dynamic response of aircraft to turbulence in order to maintain an adequate level of safety. The more arbitrary discrete gust approach does not permit the realistic representation of the continuous nature of atmospheric turbulence necessary to take into account airplane response characteristics in a rational manner."

Thus, it can be seen that the past, present and future VGH programs provide valuable turbulence data to the aircraft designer. However, due to the inherent limitations, these data cannot be applied with confidence to the design of aerospace vehicles of advanced design.

GUST DATA AVAILABLE - POWER SPECTRAL DENSITY (PDS)

A typical airborne gust measuring system required to resolve three axis, continuous true gust velocities is much more complex than the VGH type. In addition to the VGH parameters, at least seventeen additional variables are recorded, including angles of attack and yaw, gyro outputs from three axes aircraft attitude and angular rate changes, ambient temperature, etc. The statistical and analytical techniques required to reduce the recorded flight data to power spectral density format are also complex and expensive.

Because of the high costs associated with continuous gust measuring programs, the amount of data available is comparatively limited. Research programs in this area have been conducted principally in the US and Great Britain. The most significant

single effort was conducted and funded by the U. S. Air Force. A specially instrumented Douglas NB-66B jet aircraft was flown on a low level gust investigation and through turbulence associated with thunderstorms at higher altitudes. Published data from these programs are as follows:

- a. "B-66 Low Level Gust Study" - Report WADD TR-60-305, 14 Volumes and 89 books, 1958-61.
- b. NB-66B High Altitude Gust Survey - Report ASD-TDR 63-145, 3 volumes, 1961-63, "Participation in National Severe Storm Project to Gather Thunderstorm Data".
- c. "B-66B Summer Low Level Gust Study" - Report ASD-TDR 62-1036, 2 Volumes, 1961-63.

The B-66 programs developed the methodology for conducting gust measurements but was hampered by low sample size (32.5 hrs.), too little meteorological variation and limited geographical coverage. Due to limited altitude performance, no spectral data is available from this program for altitudes above 40,000 ft.

To provide continuous turbulence data in the 40-70,000 ft. altitude regime, the AF sponsored the High Altitude Clear Air Turbulence (HICAT) Program. A fully gust instrumented Lockheed U-2 flew HICAT search missions during Phase I (April - July 1964) at Edwards AFB, California, Patrick AFB, Florida, and Ramey AFB, Puerto Rico. During 60.6 flight hours, 6.1 hours of turbulence data was recorded. Phase I was concluded 30 November 1964 to improve the instrumentation in order to provide the capability for measuring longer wavelength turbulence data. An interim report covering Phase I data will be published in June 1965.

INSTRUMENTATION SYSTEMS

The USAF has initiated programs to provide the technology to define and measure clear air turbulence and gather sufficient data for the development of valid structural data. These programs will develop improved techniques for making continuous true gust velocity measurements at all altitude levels from sea level to 200,000 ft. To obtain this broad coverage of turbulence data many technical factors must be taken into consideration.

An airborne turbulence measuring system capable of resolving continuous true gust velocities about all three axes constitutes a complex design problem. Many compromises are made to evolve an optimum system to meet specific design criteria requirements, such as:

- a. Minimum and maximum wavelength resolution required.
- b. Minimum and maximum gust velocities to be resolved.
- c. Amount of data required.
- d. Accuracy limits for overall system.
- e. Recorded data in format suitable for design criteria analyses methods.

Once given such a set of specifications the engineer's next step is the choice of a suitable test vehicle. Problems associated with this selection are:

- a. Availability and priority
- b. Vehicle sustained level flight speed capability. The sensitivity of a given airborne instrumentation system is directly proportional to test vehicle velocity.
- c. Maintenance and support requirements.
- d. Availability of properly qualified flight crews.
- e. Volume, weight and electrical power available for instrumentation system.

Once the criteria requirements and test vehicle performance characteristics are known, the actual design of the instrumentation system can begin. Based on vehicle response data and estimated gust intensities, sensors can be selected with a range of values which will insure maximum signal output. The choice of an optimum data recorder should be based, in part at least, on the following considerations:

- a. Maximum record time required.
- b. Maximum allowable noise level associated with selected tape speed.
- c. Type of multiplexing system desired.
- d. Analog vs digital system.
- e. Volume, weight and power requirements.

A complete system should have the following qualities:

- a. Quick checkout capability
- b. Ground and in-flight calibration capability.
- c. Low overall noise level.
- d. Ease of maintenance.

A typical gust measuring system measures at least 20 variables which are related to appropriate sensors as shown, Figure 4. The selection of sensors of appropriate range and sensitivity must be based on anticipated gust intensities as well as aircraft velocity and response characteristics. Typical instrumentation list are shown for the Douglas B-66 Low Level Gust Program, Figure 5 and for the HICAT U-2 Figure 6.

The recording systems are usually either narrow band frequency modulation (NBFM) or digital such as Pulse Code Modulation (PCM). Both systems are being used in Air Force turbulence programs.

The choice of an optimum system depends on various factors such as:

- a. Amount of data to be generated.
- b. Overall system accuracy required.
- c. Program cost.
- d. Reliability and Maintenance.

USAF CAT PROGRAMS

Because of the requirements for the use of different test vehicles and instrumentation for each altitude range the USAF ALLCAT program is divided into projects as follows:

a. Low-Low Altitude Clear Air Turbulence Project (LO-LOCAT)-0-1000 ft. altitude range. Four (4) turbulence instrumented USAF C-131B aircraft will be operated for a full year from four Air Force Bases, namely Edwards, Lowry, Giffiss and McConnell. Each aircraft will be flown over a specific low level route in the vicinity of its assigned base. These low level routes include all types of terrain from water to

high mountains. Three flights a day (dawn, mid-morning and mid-afternoon) will be made, on the average, every other day. The flights will be approximately one hour duration with a minimum of thirty minutes of usable turbulence data. This will result in approximately 450 hours flying time per aircraft during the year. Two altitudes, tentatively 250 feet and 750 feet, will be flown in the contour (terrain following) mode. During the second year, the instrumented C-131B aircraft and a turbulence instrumented aircraft capable of higher velocity and maneuverability will be operated in a similar manner over other types of terrain not covered the first year.

b. Low Altitude Clear Air Turbulence Project (LOCAT)-1000-20,000 ft. altitude range. A turbulence instrumented high speed aircraft, such as an F-4C, F-111 or F-106, will be flown for two years from various geographical locations, such as Hanscom, Wright-Patterson, Shaw, Dow, Edwards, Mt. Home, Pope Air Force Bases, etc., during the seasons of the year of predicted peak turbulence intensity. The aircraft will be flown both subsonically and supersonically in level flight over various types of terrain, near the Mid-Latitude Jet Stream, and through other predicted turbulent areas, covering the altitude range of 1,000 to 20,000 feet. On the average, three flights per week will be made. The flights will be approximately two hours duration with a minimum of thirty minutes of usable turbulence data.

c. Medium Altitude Clear Air Turbulence Project (MEDCAT)-20,000 - 40,000 altitude range. A turbulence instrumented high speed aircraft, such as an F-4C, F-111 or F-106, will be flown for two years from various geographical locations such as Hanscom, Wright-Patterson, Shaw, Dow, Edwards, Mt. Home, Pope Air Force Bases, etc., during the seasons of the year of predicted peak turbulence intensity. The aircraft will be flown both subsonically and supersonically in level flight over various types of terrain, in and near the Mid-Latitude Jet Stream and through other predicted turbulent areas, covering the altitude range of 20,000 to 40,000 feet. Some of the flights will be made in conjunction with the LOCAT and HICAT Projects. On the

average, three flights per week will be made. The flights will be approximately two hours duration with a minimum of twenty minutes of usable turbulence data.

d. High Altitude Clear Air Turbulence Project (HICAT)-40,000 - 70,000 ft. altitude range. A turbulence instrumented U-2 aircraft will be flown for two years from various geographical locations such as Hanscom, Edwards, Ramey, Patrick, Mt. Home, Eielson and Hickam AF Bases and in the New Zealand/Australia area. These test flights will be conducted during the seasons of the year of predicted peak turbulence intensity. The aircraft will be flown in level flight through the various Jet Streams (Sub-Tropical, Mid-Latitude, Canadian Night Polar, etc.) and other predicted turbulent areas, covering the altitude range of 40,000 to 70,000 feet. On the average, two to three flights per week will be made. The flights will be approximately four hours duration with a minimum of twenty-five minutes of usable turbulence data.

e. High-High Altitude Clear Air Turbulence Project (HH-HICAT)-70,000 - 200,000 ft. altitude range. Specially turbulence instrumented research vehicles, an SR-71 aircraft for the lower portion of the altitude range and a special constructed vehicle for the upper portions of the altitude range, will be utilized. These vehicles will be flown in level flight through areas of predicted turbulence, covering the altitude range of 70,000 to 200,000 feet.

The overall flight test schedule for the ALICAT Program is shown in Figure 7.

METEOROLOGICAL SUPPORT DATA

The effects of meteorological conditions on atmospheric turbulence are important factors in any turbulence investigation. Wind speeds and directions, wind shear, atmospheric stability, temperature, temperature gradient, cloud coverages, times of day and general air mass movement data will be required for each test flight. These data, in addition to terrain profiles, constitute the variables with which the turbulence spectra data will be correlated.

FUTURE DEVELOPMENTS OF PLANS

Theoretically, gust waves range in length from zero to infinity, spectral shapes vary from one turbulence phenomenon to another and gust velocities extend from zero to several hundred ft/sec. Data covering the practical limits of the turbulence spectrum is necessary to establish rational design criteria for future aerospace vehicles. Accomplishing this task with today's subsonic aircraft requires high resolution instrumentation systems.

Utilizing subsonic test aircraft, long wavelengths (up to 100,000 ft) corresponds to very low frequencies in order of 0.01 cycles/sec. Detecting the vertical accelerations associated with such a long, low frequency wave, assuming a minimum gust velocity of 1 ft/sec. double amplitude, requires a resolution of better than one milli-g. In terms of angular measurement in the pitch plane, a gyro is required to resolve angles of three minutes of arc (0.05 degrees) or better. Correspondingly low drift rates are required for the instrumentation used to make these measurements.

To achieve this level of accuracy, research efforts in many specialized areas are being carried on by the AF. Typical of these projects are:

a. In-house AF project to develop a differential pressure gust probe capable of accepting transducers of sufficient pressure range to cover the velocity, altitude spectrum of various test aircraft.

b. Continued development of fixed vane gust probe which was initially tested during Phase I of the HICAT Program.

c. Installation of force balance type accelerometers on an inertial platform to provide precise aircraft attitude measurements and accelerometer outputs.

d. Use of Schuler tuning to further increase the accuracy of the inertial platform.

e. Develop numerical filtering techniques and appropriate FORTRAN subroutines for use with IBM 7094 computer. This filtering technique provides sharp cutoff characteristics not attainable with electrical/analog filters.

These development projects are being integrated into the overall USAF ALICAT program to completely define atmospheric turbulence as it affects the aircraft structure.

SUMMARY

Within the next 20 years, US military and civil aerospace vehicles will be flying at supersonic velocities at all altitudes within the sensible atmosphere on a global basis. Assuming a logical, orderly increase in vehicle performance, similar to that achieved in the past 20 years, leads to this conclusion. Compared to today's aircraft these aerospace vehicles will be much heavier, larger, and of radically different configuration.

Present day meteorological studies indicate the presence of turbulence at every operational altitude level for any winged vehicle. Since the effects of turbulence on an aircraft have clearly proven to be swift, uncertain and sometimes violent, every effort must be made to define atmospheric turbulence in terms of its input to a vehicle structure.

The most promising technique capable of meeting this requirement is continuous true gust velocity measurements reduced to power spectral density format. The instrumentation systems and data analysis methods required are inherently complex and expensive. It is also clear that the atmosphere as a data sample is infinite in size and time.

As in the past, the U. S. Air Force has taken the lead and initiated programs to acquire turbulence data to meet specific design criteria requirements. However, no single agency can solve all the problems caused by CAT. The Air Force turbulence measuring programs must be done in conjunction with other areas of basic meteorological research. Only by the concerted efforts of every interested agency both private industry and government can a level of technology be reached which will establish proper design criteria and flight operational procedures. Only then can we be assured that our future aerospace vehicles will be flying in an atmosphere which has been carefully studied and measured to insure safe, smooth high speed flight.

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POWER SPECTRAL DENSITY PLOT

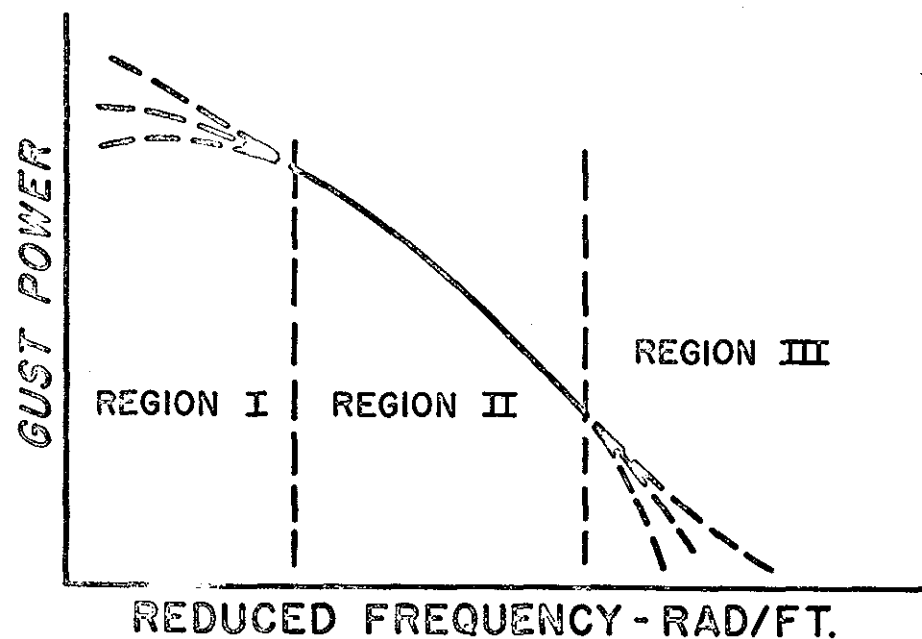
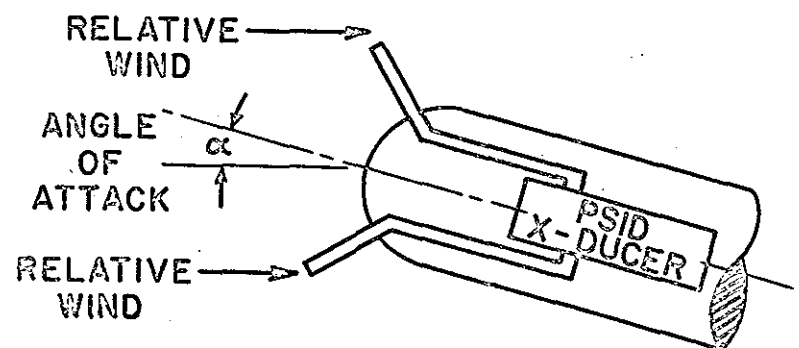
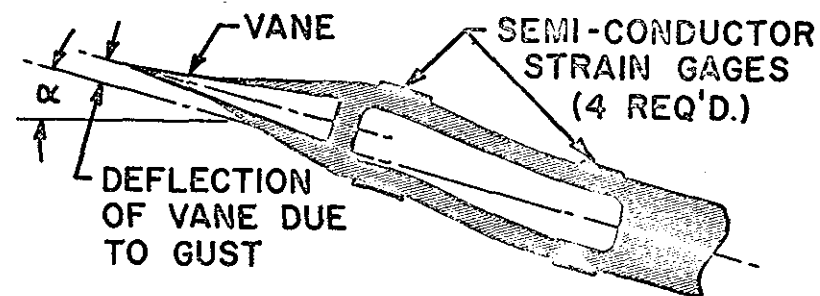


FIGURE 1

GUST PROBES



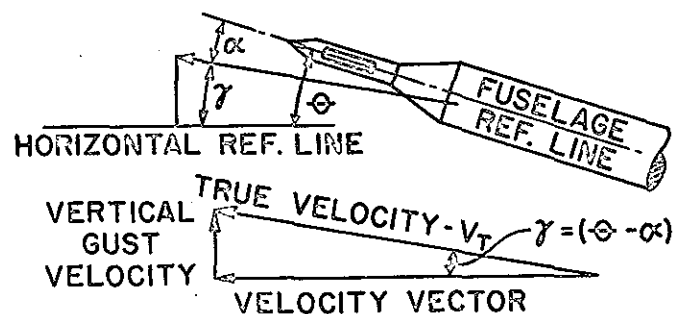
(A) DIFFERENTIAL PRESSURE PROBE



(B) FIXED VANE GUST PROBE

FIGURE 2

BASIC TRUE GUST VELOCITY EQUATION



$$\text{TRUE GUST VELOCITY} - U_V = W_N - (V_T)(\phi - \alpha)$$

W_N = VERTICAL VELOCITY OF PROBE IN SPACE

V_T = TRUE VELOCITY

ϕ = PITCH ATTITUDE

α = ANGLE OF ATTACK

γ = FLIGHT PATH ANGLE OF PROBE

FIGURE 3

TRANSDUCERS AND RELATED GUSTS

	VERTICAL GUST COMPONENT	LATERAL GUST COMPONENT	FORWARD GUST COMPONENT
ANGLE OF ATTACK TRANSDUCER	X		
ANGLE OF SIDESLIP TRANSDUCER		X	
AIRSPEED TRANSDUCER	TAS	TAS	FOR ΔV_T
VERTICAL ACCELEROMETER	X		
LATERAL ACCELEROMETER		X	
LONGITUDINAL ACCELEROMETER			X
PITCH ATTITUDE GYRO	X		
ROLL ATTITUDE GYRO		X	
3 AXIS RATE GYRO	X	X	X
OUTSIDE AIR TEMPERATURE	TAS	TAS	TAS
ALTITUDE TRANSDUCER	TAS	TAS	X

FIGURE 4

INSTRUMENT	TRANSDUCER	LOCATION	MAKER	TYPE	RANGE	FULL SCALE USED
Differential Pressure (Angle of attack)	Boom	Statham	PM231TC	+2.5-350	± 2.5 psid	± 1.1 psid
Differential Pressure (Angle of sideslip)	Boom	Statham	PM231TC	+2.5-350	± 2.5 psid	± 1.2 psid
Differential Pressure (Airspeed-Low)	Boom	Statham	PM231TC	+2.5-350	± 2.5 psid	0-2.5 psid
Differential Pressure (Airspeed-High)	Boom	Statham	PM131TC	+5.0-350	± 5 psid	0-5.0 psid
Accelerometer (Vertical)	C. G.	Statham	AJ43-2	-350	± 2 g	0-2 g
Accelerometer (Vertical)	C. G.	Statham	A 43-6	-350	± 6 g	± 3.5 g
Accelerometer (Vertical)	Radome	Statham	AJ43-2	-350	± 2 g	0-2 g = 1.8 g
Accelerometer (Lateral)	Radome	Statham	AJ43-0.5	-350	± 0.5 g	± 0.5 g
Accelerometer (Longitudinal)	Radome	Statham	AJ43-0.6	-350	± 0.6 g	± 0.6 g
Rate Gyro (Pitch)	Radome	Giannini	36128DS	-2-15R	$\pm 15^\circ/\text{Sec}$	$\pm 6.1^\circ/\text{Sec}$
Rate Gyro (Yaw)	Radome	Giannini	36128DS	-2-15R	$\pm 15^\circ/\text{Sec}$	$\pm 6.2^\circ/\text{Sec}$
Attitude Gyro (Pitch)	Radome	Minneapolis Honeywell	XJG7044A	-35	$\pm 43^\circ$	$\pm 9.6^\circ$
Attitude Gyro (Roll)	Radome	Minneapolis Honeywell	XJG7044A	-35	$\pm 180^\circ$	$\pm 17.2^\circ$
Radar Altimeter	Bomb Bay	Raytheon	AN/APN-22		0-20,000	0-10,000 Ft
Pressure Transducer (Altitude)	Radome	Statham	PA177TC	-15-350	0-15 psia	15-6.5 psia
Temperature Probe (OAT)	Nose Gear Door	Douglas	Low Lag			0-60°C

FIGURE 5, B-66 INSTRUMENTATION LIST

INSTRUMENTATION LIST - HICAT RESEARCH AIRCRAFT

Parameter	Manufacturer	Model	Serial No.	Instr. Range	Bendix Osc. No.*	Location & Remarks
1. Alpha-Vane Normal Force	Lockheed	2	4	± 12 LB	B-1162	Gust Probe
2. Beta-Vane	Lockheed	2	5	± 12 LB	8V321	Gust Probe
3. Indicated Airspeed	Statham	PI283TC	722	0-1 PSI	B-873	Gust Probe
4. Boom Lateral Acceleration	Statham	AJ-43-350	10441	± 3 g	91421	Gust Probe
5. Boom Normal	Statham	AJ-43-350	10440	± 3 g	B-1161	Gust Probe
6. Pitch Rate	Humphrey	RQ-02-0501	AP60-H24	57 $^\circ/\text{sec}$	B-253	Nose Compartment
7. Yaw Rate	Humphrey	RQ-02-0501	AP60-H24	57 $^\circ/\text{sec}$	8X318	
8. Roll Rate	Humphrey	RQ-02-0501	AP60-H24	57 $^\circ/\text{sec}$	B-1163	
9. Outside Air Temperature	Lockheed	None	210P-17*	$\pm 130^\circ$ F	B-1146	A/C Nose
10. Pressure Altitude	Rosemount	800R544	101	0-4 PSI	None	Nose Compartment
11. C.G. Longitudinal Acceleration	Statham	A26-1-350	184	± 1 g	8V327	Fuselage
12. C.G. Normal Acceleration	B. & F.	IF-5-15-120	577	± 5 g	8X322	Fuselage

FIGURE 6

* Model TOR-7

x Lockheed Key No.

INSTRUMENTATION LIST - HICAT RESEARCH AIRCRAFT

Page Two

Parameter	Manufacturer	Model	Serial No.	Instr. Range	Bendix Osc. No.	Location & Remarks
13. Right Wing Nodal Acceleration	Statham	A 43-4-350	350	+ 4 g	8V323	Wing Sta. 163 ±
14. Left Wing Nodal Acceleration	Statham	A 43-2-350	215	+ 2 g	B-871	Wing Sta. 163 ±
15. Elevator Position	Spectral	100STD	—	+ 5000 (355°)	B-1265	Elevator Hinge
16. Rudder Position	Spectral	100STD	—	+ 5000 (355°)	B-1167	Rudder Hinge
17. Aileron Position	Spectral	100STD	—	+ 5000 (355°)	B-1098	Aileron Hinge
18. Pitch Angle	Minneapolis-Honeywell	JG7044A40	M-169	+ 80°	8X-317	Nose Compartment
19. Yaw Angle	Minneapolis-Honeywell	—	—	360°	B-508	Nose Compartment
20. Roll Angle	Minneapolis-Honeywell	JG7044A40	M-169	360°	8X377	Nose Compartment

FIGURE 6 (CONT'D)

± Perpendicular to aircraft water line

USAF CAT PROGRAMS (PROPOSED)

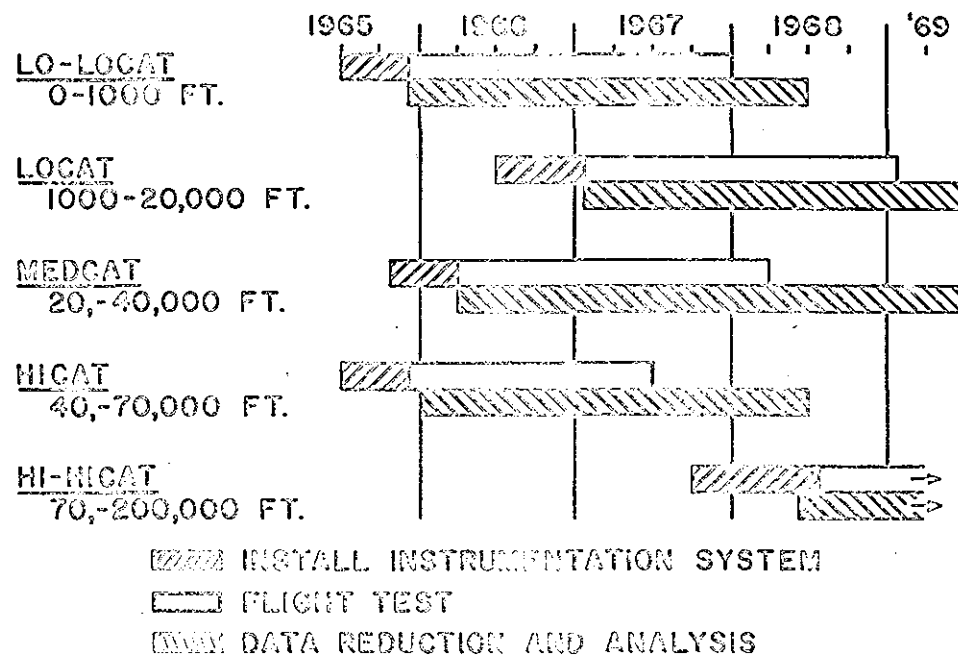


FIGURE 7